

Advancing scanning probe microscopy simulations: A decade of development in probe-particle models

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Since its inception more than a decade ago [1], the probe particle model has become an important tool for simulating high-resolution SPM images obtained with flexible tip apices, shedding light on the origin of the contrast patterns observed in experiments. Our recent review [2] discusses the development of the probe particle simulations over the years, with emphasis on the open source ppafm package for AFM simulations [3]. Starting from a simple model utilizing a Lennard-Jones force field, ppafm has expanded to include various force field models for electrostatic and Pauli interactions utilizing data from ab initio calculations, as well as other simulation modes including STM and KPFM. The GPU acceleration has sped up the simulations by 2-3 orders of magnitude, allowing for development of machine learning tools for SPM image interpretation [4]. Finishing with a graphical user interface making the package more user friendly and easier to approach. At the very end, I will shortly introduce my current project on recovering surface reconstruction with a help of machine learning and scanning probe microscopy simulations.

[1] P. Hapala et al., *Phys. Rev. B* **90**, 085421 (2014).

[2] N. Oinonen et al, *Comput. Phys. Commun.* **305**, 109341 (2024).

[3] P. Hapala et al, *ppafm repository*: <https://github.com/Probe-Particle/ppafm> **0**, 0 (2026).

[4] B. Alldritt et al., *Sci. Adv.* **6**, eaay6913 (2020).